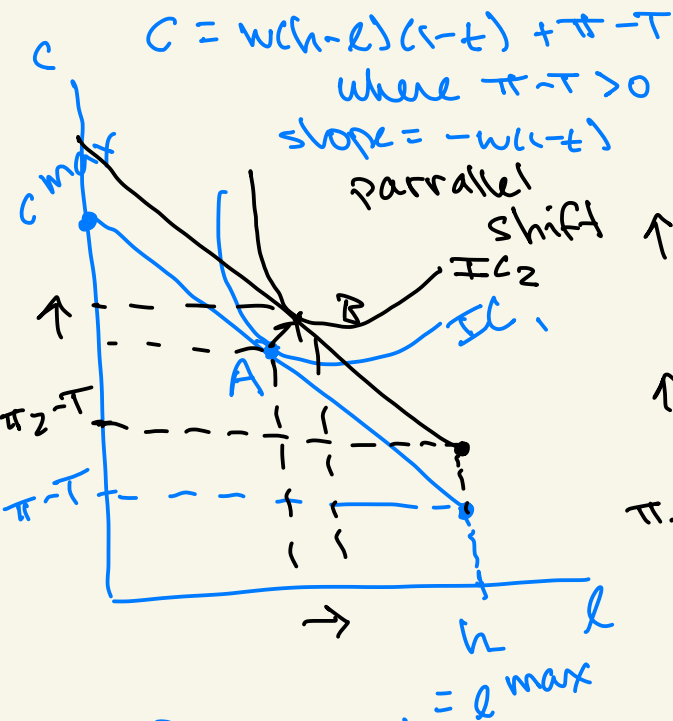


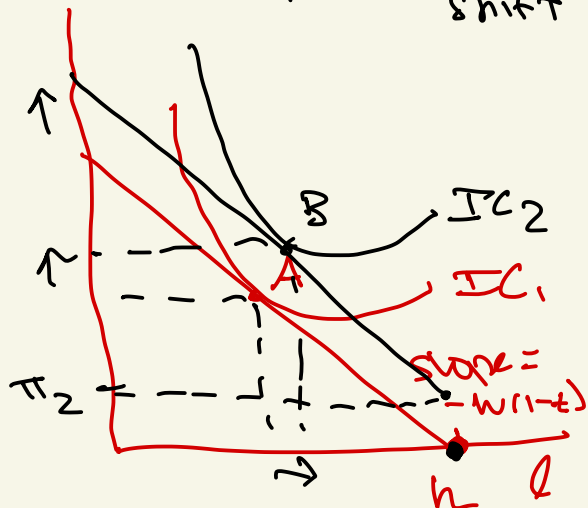



$$MRS = w(1-t)$$

$$\frac{\Delta C}{\Delta l} = \frac{MU(l)}{MU(C)} = w(1-t) \text{ at A}$$

what if $\pi \uparrow$
 C, l normal goods
 so $C \uparrow$ $l \uparrow$ ($N \downarrow$)
 $U \uparrow$ (higher IC)

where $\pi - T = 0$
 parallel shift



$$MRS = w(1-t)$$

$$\frac{MU(l)}{MU(C)} = w(1-t) \text{ at A}$$

qualitative
 d's

"Pure Income Effect"

(no Δ slope)

* This is "partial equilibrium" analysis.

unanswered / unmodeled questions:

Where did π come from?

What determines w ?

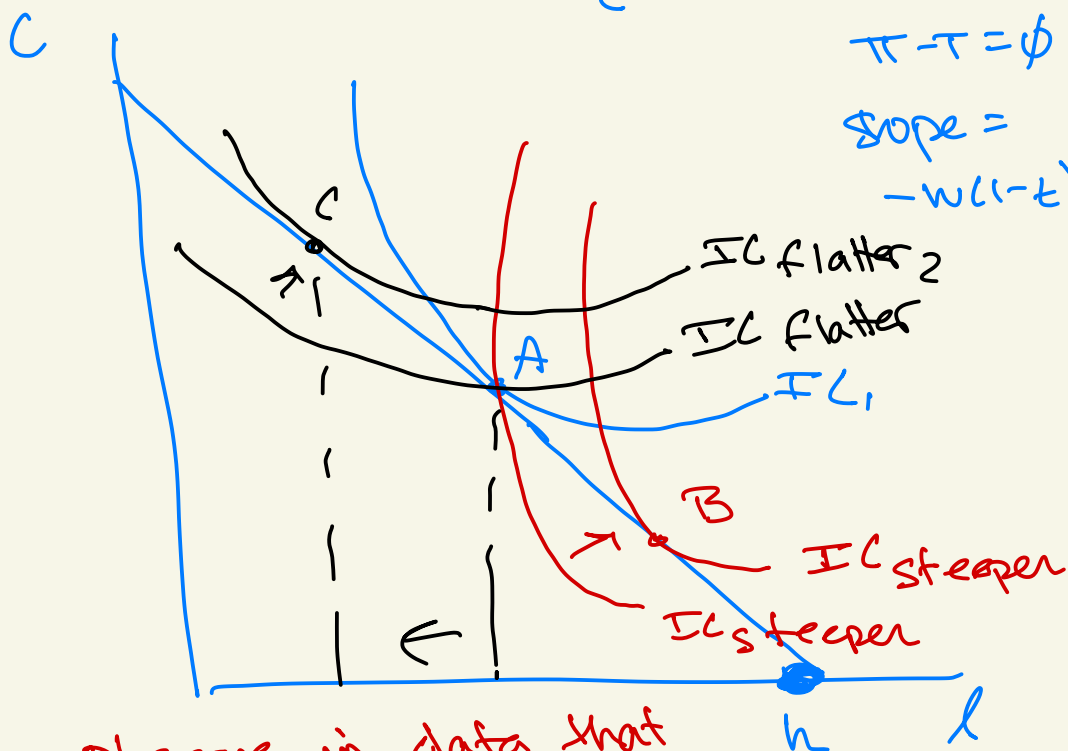
$\Delta N \approx 20$ but no impact
on wages

Example #2

$$C = w(h-l)(1-t) + \pi - T$$

$$\pi - T = \phi$$

$$\text{slope} = -w(1-t)$$



Observe in data that women are working more $\rightarrow$ how could I model this as a result of a Δ in preferences (IC)

at B: $l \uparrow$ $N \downarrow$ } reject

at C: $l \downarrow$ $N \uparrow$ } okay

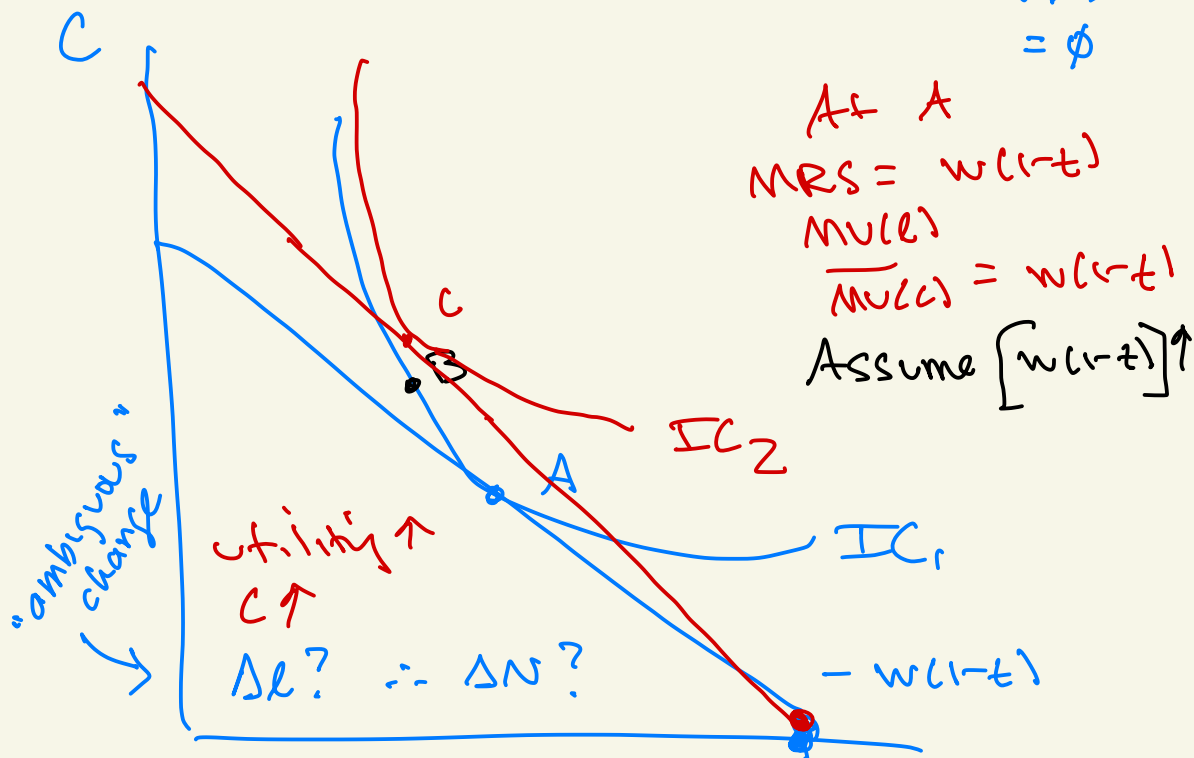
flatter IC means slope lower

either like l
less or C
more

$$\begin{matrix} \downarrow \frac{MU(l)}{MU(C)} \\ \uparrow \frac{MU(l)}{MU(C)} \end{matrix} \quad \downarrow$$

Example #3

$$C = w(1-t)(h-l) + \underbrace{\pi - T}_{= \phi}$$



At A
 $MRS = w(1-t)$
 $\frac{MU(C)}{MU(L)} = w(1-t)$
 Assume $[w(1-t)] \uparrow$

New equilibrium where

$$\uparrow \frac{MU(L)}{MU(C)} = [w(1-t)] \uparrow$$

substitution effect

if $\underline{l \downarrow}$ or $\underline{C \uparrow}$
 $A \rightarrow B$ $N \uparrow$

Both happen along IC₁

Budget constraint ↑ = Income effect
 $C \uparrow$ $\underline{l \uparrow}$
 normal goods

Representative Firm

Firm is a profit-maximizer
(wage-taker)

$$\pi = \text{profits} \\ = \text{output} - \text{wage bill}$$

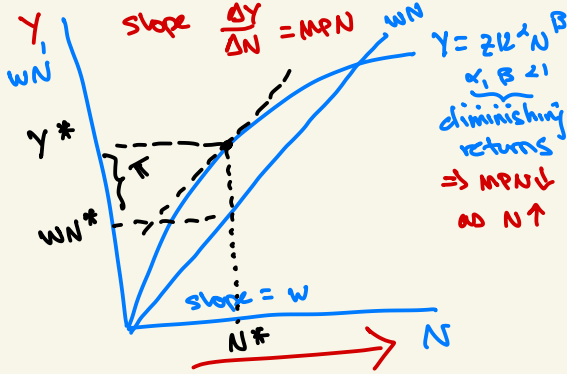
$$\pi = ZF(K, N) - WN$$

at π -max, marginal profits = 0

$$\frac{\partial \pi}{\partial N} = 0 = \frac{\partial Y}{\partial N} - \frac{\partial (WN)}{\partial N}$$

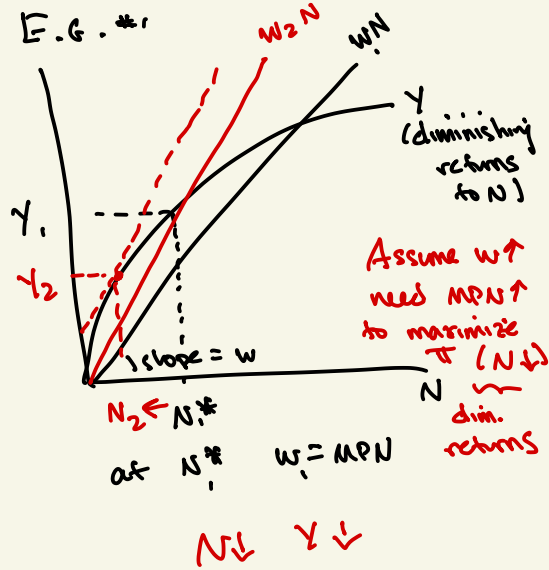
$$0 = \underbrace{MPN}_{\text{marginal productivity of } N} - w$$

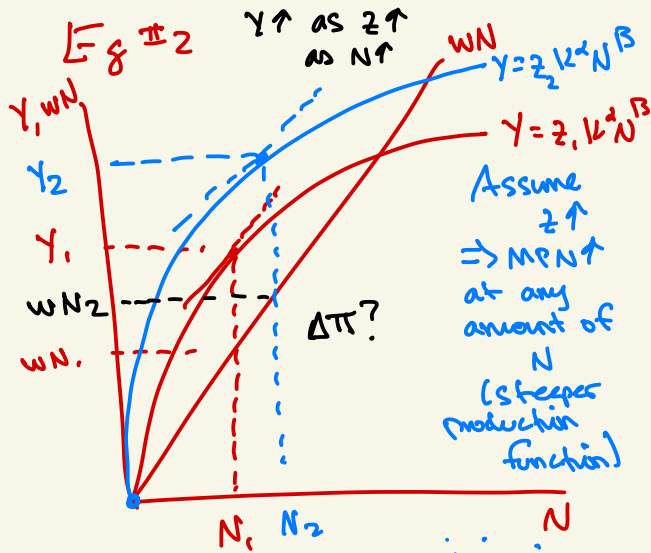
$$w = MPN \quad \left. \begin{array}{l} \text{choose } N \\ \text{to get} \end{array} \right\} MPN = w$$



$$\pi = Y - WN$$

choose N such that
 the gap between Y and WN
 is the biggest it can be
 where $MPN = w$
 where have
 same slope





$w = MPN$ in equilibrium

$\underbrace{\quad}_{\text{no } \Delta}$ $MPN \uparrow$ as $z \uparrow$

diminishing returns } $MPN \downarrow$ again to equal w as $N \uparrow$